# *M-Mobile and other 'superfluous' forms of nasal onsets in Kartvelian*

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Kartvelian languages are famous for their exuberant consonant onsets clusters that violate sonority sequencing principles, notably nasal onsets. In reality, there is substantial variation amongst them to what extent they exhibit such onsets. This paper will propose that variation amongst nasal onsets in the family derives not only from retention from the protolanguage but from separate processes of nasal onset deletion in Megrelian, nasal onset epenthesis in Laz, and a process of morphophonological reanalysis called 'M-mobile' in Georgian.

Amongst the first things anyone studying Georgian learns about the language is its baroque consonant clusters not found in more familiar western languages. Words such as მკერდი *mk'erdi* 'chest', მზე *mze* 'sun', მთა *mta* 'mountain' stand out in their syllabic profile because of the way in which they violate common crosslinguistic norms of sonority sequencing principles in which words reach a sonority peak at the syllabic nucleus and decline in sonority before and after. Consider an English word like 'grudged' and a Georgian word like მწვრთნელი *mc'vrtneli* 'trainer':

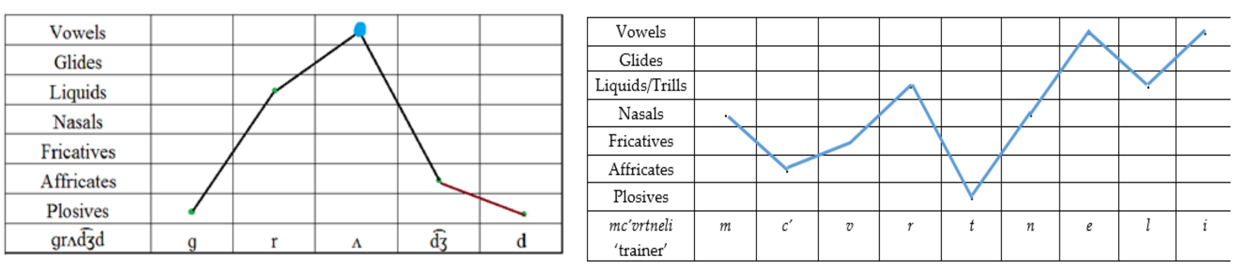


Figure 1. Sonority profiles of the English word 'grudge' and the Georgian word მწვრთნელი*mc'vrtneli*'trainer'

Such radically different sonority profiles in syllable structure suggest that Georgian speakers operate under different phonological constraints than speakers of English and many other languages do: nasal onsets are valid in clusters even when they violate a standard sonority contour. Similar violations of typological norms of sonority are also found across the family too in Megrelian, Laz and Svan, though each language responds to the constraint in different ways. In the following sections, we will discuss these differences, and see that while Megrelian and Svan tend to eliminate nasal onset clusters by deletion or vocalic epenthesis, Georgian and Laz actually increase the number of nasal onsets relative to their reconstructed ancestral forms by nasal epenthesis or by a process of morphophonological rebracketing. In other words, not all nasal onsets in Kartvelian are inherited; many are secondary, and at least one major class is due to morphophonological reanalysis triggered by demonstrative-final nasals that I dub 'M-Mobile'.

## §1 The behavior of nasal onsets across Kartvelian

To better understand how nasal onsets function across Kartvelian, it is important to examine them as cognate sets with differential sets of phonotactic properties. Consider the following sets in Table 1:

| | **Georgian** | **Megrelian** | **Laz** | **Svan** | **Source** |
|---|---|---|---|---|---|
| a. **m(i)zʲe* 'sun' | mze | mža, bža | mža, bža | miž, məž | Fähnrich (2008: 289) |
| b. **gweł-* 'snake' (GZ)[1] | gveli | gveri, ngveri | mgveri | - | Fähnrich (2008: 105) |
| c. **mcʲ'erʲ-* 'fly' | mc'eri | č'andi, č'anji | m'čaji | mēr | Fähnrich (2008: 648) |
| d. **(m)c'aw-* 'otter' | mc'avi | c'vinori | c'vinari | c'iwēr | Fähnrich (2008: 613) |
| e. **mt'wer-* 'dust' (GZ) | mt'veri | t'veri | mt'veri | - | Klimov (1998: 126) |

**Table 1. Kartvelian cognate sets with nasal onsets.**

[1] In this and later tables, the designation '(GZ)' indicates that a form can only be reconstructed to the Georgian-Zan intermediary stage, after Svan broke off from common Kartvelian. For more information on the reconstruction of specific features of Proto-Kartvelian or Proto-Georgian-Zan, see Wier (

In some cases, as in (a) **m(i)z$^{j}$e* 'sun', we see cognate sets in which all forms have a nasal onset, albeit not always stably or constituting a cluster *per se*: notably in Zan languages the nasal onset tends to lose its nasality, and in Svan it is a simple onset. In other cognate sets, such as those belong to class (b) **gweł-* 'snake', both of the Zan languages bear a nasal onset, while they are not found in the sister languages. In words belonging to class (c) **mc$^{j}$er$^{j}$-* 'fly', all cognates bear a nasal onset, except for Megrelian, while in words belonging to class (d) *(*m)c'aw-* 'otter', only the Georgian cognate bears a nasal onset. The last class of (e) **mt'wer-* 'dust' represents words in which the Georgian and Laz forms bear a nasal onset, but such an onset is lacking in the other sister languages. So we see that there is in fact considerable variation in the behavior of nasal onsets across the family.

Complicating this picture is the fact that in many cases one or more of the languages lacks a cognate, especially the phylogenetic outlier Svan. This makes the analytical significance of some sets less than totally clear. Traditional reconstructions of Kartvelian (e.g. Schmidt 1962; Gamkrelidze & Machavariani 1965; Klimov 1998; Fähnrich & Sarjveladze 1990) do not always make explicit which nasals they posit to be conservative retentions from the protolanguage, and which they posit to be innovations in one or more daughter branches, and why. For example in Fähnrich's *Kartwelisches Etymologisches Wörterbuch* (Fähnrich 2007: 303), he provides the following cognate set:

(1) **mc'q'aw-* 'cherry laurel'

| | |
|---|---|
| (a) Georgian: | mc'q'avi |
| (b) Megrelian: | c'q'ei, c'q'i, c'q'oli |
| (c) Laz: | c'u, mc'u, mc'k'oli, mc'k'o |
| (d) Svan: | c'q'aw, c'qew |

Fähnrich reconstructs this cognate set (analytically the same as Group D or E in Table 1) with a nasal onset even though three of the four daughter languages lack it in some or all dialects, and the one that consistently possesses the nasal onset, Georgian, is usually not considered to be the phylogenetic outlier. This is presumably because, in Georgian, though not the sister languages, there is no obvious systematic phonological sound-law that epenthesizes nasals in word-initial position. In what follows, to account for these variations, I will suggest that the nasal onsets represented in these cognate sets have four distinct historical sources:

(2) a. The retention of nasals onsets from the protolanguage (that is, the null hypothesis);

b. The deletion of nasal onsets in Megrelian;

c. Nasal epenthesis in accented syllables in Zan languages, with differential effects in Megrelian (epenthesis after the syllable nucleus) and Laz (epenthesis into either the syllabic nucleus, before the initial onset, or both)

d. Cases of 'M-mobile': the reanalysis and rebracketing of word-boundaries in Georgian and Laz

The apparent irregularity of these cognate sets is thus only superficial: the different distributions of nasal onsets reflect four distinct historical pathways, each of which can be evaluated against the comparative and dialectal evidence to which we now turn.

## §2 Nasal onsets and their reflexes across Kartvelian

### 2.1 Retention from the Protolanguage

It is clear from the outset that many cases of complex onsets with an initial nasal must have been part of the linguistic inheritance from earlier stages of the language family. We find numerous cognate sets in which all attested daughter languages bear a nasal onset, at least in one dialect form. Thus in Table 2 we find forms like Georgian *mze* 'sun', Megrelian *mža*, Laz *mža*, and Svan *miž*. In the case of 'moth' (Table 2d), we see evidence that such onsets were probably a feature of the Proto-Kartvelian protolanguage, since Svan is usually taken to be the phylogenetic outlier.

| Gloss | Georgian | Megrelian | Laz | Svan | Source |
|---|---|---|---|---|---|
| a. **m(i)zʲe* 'sun' | *mze* | *mža, bža* | *mža, bža* | *miž, məž* | Fähnrich (2008: 289) |
| b. **m(i)zʲwar-* 'sunny' | *mzore* | - | *(m)žora, bžora* | *məžä:r* | Klimov (1998: 121) |
| c. **mxar-* 'midmorning meal' | *sa-mxar-i* | - | - | *məxär* 'morning' | Fähnrich (2008: 306) |
| d. **mğ–* 'moth' | *mğili (OG)* | - | - | *mğul-* | Klimov (1998: 127) |

**Table 2. Nasal onsets retained from ancestral forms of the language family**

In some of the individual languages, especially including Georgian, initial *m*-onsets often bear a specific morphological function: they constitute an agentive prefix that attaches immediately to verb roots to create participial nominalizations, usually in conjunction with other suffixal material such as thematic suffixes *–eb/-av* and participial suffixes *–il/-el/-al*. Thus we see forms like those in Table 3, illustrating agentive deverbal nominalizations:

**Table 3. Georgian *M*-agentive nominalizations with nasal complex onset**

| Georgian agentive nominalizations | | | |
|---|---|---|---|
| *m-brdzen-eb-el-i* | commander | *m-bar-av-i* | digger |
| *m-tes-av-i* | sower, planter | *m-basr-av-i* | cutter, hewer |
| *m-dev-ar-i* | persecutor<br>(lit. seeker) | *m-zom-el-i* | surveyor<br>(lit. measurer) |
| *m-txovn-el-i* | applicant<br>(lit. requester) | *m-ban-el-i* | bath attendant<br>(lit. bather) |

Did Proto-Kartvelian have such agentive nominalizations? This seems likely, though we find few cognate sets that can directly prove this. In many cases we find such are features of the later Georgian-Zan intermediary stage, as in Table 4:

| Gloss | Georgian | Megrelian | Laz | Svan | Source |
|---|---|---|---|---|---|
| **m-k'al-e* 'short, lacking'(GZ) | *mk'le* (OG), *mok'le* | *k'ule* 'deprived' | *mk'ule* | - | Klimov (1998: 123) |
| **m-k'wax-e* 'unripe, sour'(GZ) | *mk'vaxe* | *k'oxa* | *k'oxa* | - | Klimov (1998: 123) |
| **m-rt-eł-* 'safe, intact, whole' | *mrteli* (OG) | *tari* | *(teli)* | *tel* | Klimov (1998: 125) |
| **m-(w)sx-al-* 'bear fruit, pear' | *msxali* | *sxuli* | *sxuli* | *wicx, icx* | Klimov (1998: 125) |

| | | | | | |
|---|---|---|---|---|---|
| _*m-s$^{j}$xw-il-_ 'stout, big'(GZ) | _msxvili_ | _šxu_ | _mčxu_ | - | Klimov (1998: 125) |
| _*m-š$^{y}$w-en-_ 'beautiful' | _(m)švenieri_ | _skvami_ | _mskva_ | _musgwen_ | Klimov (1998: 128) |
| _*m-š$^{y}$w-e_ 'child' | _-mšo_ (OG) | _skua_ | - | _(m)sgey_ | Klimov (1998: 128) |
| _*m-dzew-al-_ 'foundling, egg'(GZ) | _mdzevali_ | _jaali_ | - | - | Klimov (1998: 130) |
| _*m-č$^{y}$'ed-el-_ 'blacksmith' | _mč'edeli_ | _č'k'adu_ | _mč'k'adu_ | _məšk'id_ | Klimov (1998: 133) |
| _*m-č$^{y}$'al-e_ 'lean, thin' (GZ) | _mč'le_ | _č'k'ola_ | - | - | Klimov (1998: 133) |

**Table 4. Reconstructed agentive nominalizations in Georgian-Zan/Kartvelian**

## 2.2 Nasal onset deletion in Megrelian

Whether or not _m_-initial words in Georgian reflect a morphological feature of word structure, some clearly preserve inherited _m_-initial sequences, as shown by their secondary effects in the phonological history of the sister languages. One of the characteristic sound changes of the Zan languages, for example, is the raising of _*a_ to _u_ in the presence of a labial consonant in the root (XX). For example, from Georgian-Zan _*bal-_ 'wild cherry', we see Georgian _bali_, Megrelian _buli_, and Laz _mbuli_; from Kartvelian _*bard-_ 'climbing plant' we see Georgian _bardi_ 'undergrowth', Megrelian _burdi_ 'blackthorn', and Svan _bärd_ 'ivy'; and from Kartvelian _*da(s$^{j}$)tw-_ 'bear' we see Georgian _datvi_, Megrelian _tunti_, Laz _mtuti_ and Svan _däšdw_. Such cases provide independent evidence that at least some Georgian _m_-initial forms cannot be attributed to later morphological reanalysis or epenthesis, but must preserve an inherited nasal onset:

| GLOSS | GEORGIAN | MEGRELIAN | LAZ | SVAN | SOURCE |
|---|---|---|---|---|---|
| **m-k'al-e* 'short, lacking'(GZ) | *mk'le* (OG), *mok'le* | *k'ule* 'deprived' | *mk'ule* | - | Klimov (1998: 123) |
| **m-tʼk'aw-el-* 'inch, span' (GZ) | *mtʼk'aveli* | *tʼk'u, tʼk'ou* | *mtʼk'u* | | Klimov (1998: 126) |
| **mq*$^{h}$*ar*$^{j}$- 'shoulder' | *mq*$^{h}$*ari* (OG) | *xuji* | *mxuji* | *məqär* 'arm up to elbow' | Klimov (1998: 134) |
| **mxal*-'edible greens' (GZ) | *mxali* | *xuli* | | | Klimov (1998: ) |
| **mč*$^{ɣ}$*'ad*- 'corn bread' (GZ) | *mč'adi* | *čʼk'idi* (< **čʼk'udi*) | *mčʼk'udi* | | Klimov (1998: ) |
| **m-(w)sx-al*- 'bear fruit,pear' | *msxali* | *sxuli* | *sxuli* | *wicx, icx* | Klimov (1998: 125) |

**Table 5. Cognate sets showing Megrelian *a > u without an initial *m*-onset**

Such forms provide evidence that, at the time the Zan sound-change was under way, there was an *m*-initial onset in these cognate sets present to trigger that sound-change of **a* > /*u*/. The absence in Megrelian of such onsets is therefore a result of a later sound-change of nasal-onset deletion in order to satisfy typological norms of sonority sequencing found in most languages. The near-complete absence of such onsets is obvious throughout the above cognate sets in Tables 1 through 5, but it also occurred in some cases where the relevant raising/rounding sound-change of **a* > /*u*/ seems not to have occurred, as Table 6:

| Gloss | Georgian | Megrelian | Laz | Svan | Source |
|---|---|---|---|---|---|
| **m-c*$^{j}$*'er-i*'fly' | *m'ceri* | *č'andi* | *mč'andi* | *mēr* | Klimov (1998: 131) |
| **mk'ward-* 'chest' | *mk'erd-* | *k'idir-* | - | *məčʷəd, mučod* | Fähnrich (2007:234) |
| **m-č*$^{y'}$*ed-el-* 'blacksmith'(GZ) | *mč'edeli* | *č'k'adu* | *mč'k'adu* | *məšk'id* | Klimov (1998: 133) |

**Table 6. Cognate sets showing Megrelian *m*-loss without vowel raising**

It seems thus to be the case that rules of vowel raising and nasal deletion only partially overlapped in the lexical items to which they were to apply, as there are forms that underwent one rule but not the other. This perhaps reflects dialect variation (some dialects underwent the raising rule while others did not), but in these cases it is unlikely to reflect different periods in which the respective rules were active, as we will see in the next section.

### 2.3 Nasal epenthesis in Zan languages

Contrary to this above pattern of nasal *deletion*, we also find various kinds of nasal *epenthesis* in the closely related Zan languages, both as onsets and into syllabic nuclei. We find this in all parts of the Megrelian and Laz lexicons, both on nouns and on verbs, both anciently inherited and in relatively recent loanwords, e.g. *tambako* 'tobacco' (hence borrowed into Georgian), and in Laz these form nasal onsets:

| Gloss | Georgian | Megrelian | Laz | Svan | Source |
|---|---|---|---|---|---|
| **gweł-* 'snake' (GZ) | *gveli* | *gveri* | *mgveri* | - | Fähnrich (2007: 105) |
| **gor-* 'roll' | *gor-* | *gor-* | *ngor-* | *gur-, gwr-* | Fähnrich (2007: 110) |
| **grc'q'-ił-* 'flea' | *grc'q'ili* (OG) | *c'q'iri* | *mc'q'iri* | *zisq'* | Fähnrich (2007: 113) |

| | | | | | |
|---|---|---|---|---|---|
| **da(s$^j$)tw-* 'bear' | *datvi* | *tunti* | *mtuti* | *däšdw* | Fähnrich (2007: 123) |
| **drek'-* 'bend' | *drek'-* | *drak'-* | *ndrak'-* | - | Klimov (1998: 42) |
| **tel-* 'press, beat' | *tel-* | *tal-* | *ntal-* | *tel- / tl-* | Fähnrich (2007: 191) |
| **txił-* 'hazelnut' | *txili* | *txiri* | *txiri, mtxiri, ntxiri* | *šdix* | Fähnrich (2007: 208) |
| **değ-* 'day' | *dğe* | *dğa* | *ndğa* | *değ-* | Fähnrich (2007: 131) |
| **txam-* 'alder tree' | *txmeli* | *txomu* | *ntxomu* | - | Fähnrich (2007: 208) |

**Table 7. Cognate sets showing Laz nasal prothesis**

In most cases such nasal epenthesis appears to be homorganic, in that it matches the place features of the following obstruent, though this is not uniformly so, as Laz *mtuti* 'bear' attests.

Though Megrelian lacks such nasal onsets, and indeed actively avoids them, both Megrelian and Laz underwent an entirely separate rule of nasal epenthesis into syllabic nuclei of accented syllables. We find this rule operating in both anciently inherited lexemes like **eks$^j$w-* 'six' and also relatively recent loans, such as Megrelian *manja* 'wrist', a borrowing from Georgian *maja*, which is itself a borrowing from Arabic مَجَسَّة *majassa* 'wrist, pulse, cuff'[2]. Because this particular form undergoes nasal epenthesis but not the Zan raising rule of **a* > *o*, we can be sure that the former postdates the latter by comparison to other loan words and the context of borrowing; see Wier (in press) for more discussion. Perhaps the most famous example of nuclear nasal epenthesis is the alternate form of the language name *Mingrelian*, which has been used by western languages since at least the 16$^{th}$-17$^{th}$ centuries. This form is attested in Laz dialects as *Mengreli*, whence it was borrowed into Italian and from there into English and Russian.

[2] The Arabic source form lost its final syllable in Georgian by reanalysis of *majassa* as *maja-sa* 'wrist-DAT.SG'.

| Gloss | Georgian | Megrelian | Laz | Svan | Source |
|---|---|---|---|---|---|
| **eks$^{j}$w-* ‘six’ | *ekvsi* | *amšvi* | *anši* | *usgwa* | Fähnrich (2007: 151) |
| **da(s$^{j}$)tw-* ‘bear’ | *datvi* | *tunti* | *tuti, mtuti* | *däšdw* | Fähnrich (2007:123) |
| ‘wrist’ (< Arabic مَجَسَّة *majassa*) | *maja* | *manja* | - | - | |
| **q$^{h}$id-* ‘bridge’ | *xidi* | *xinji* | *xinji* | - | Fähnrich (2007:706) |
| **k’utx-* ‘corner’ | *k’utxe* | *k’untxu* | *k’untxu* | - | Fähnrich (2007: 257) |
| **mak’e-* ‘pregnant’ | *mak’e* | *mok’e, monk’e* | *monk’e* | - | Fähnrich (2007: 278) |

**Table 8. Cognate sets with nasal epenthesis in accented syllables in Zan languages**

What could motivate such a change? Recent work by Nevins and Pinheiro (2019) has argued that a similar kind of nuclear nasal epenthesis occurs in modern Brazilian Portuguese. This language, which already bears a robust contrast of nasalized versus non-nasalized vowels, appears to use vocalic nasalization as a means to foreground or raise the prominence of stressed syllables, as in **Table 9**. In these cases, and unlike Zan, the epenthesis is not of a nasal segment with full oral closure, but rather of nasality on one or more of the unstressed vowels. But as they note, this is part of the same functional process:

> ‘[W]e claim that spontaneous nasalization is a strategy to perceptually enhance prominence, and tends towards particular vowels in particular syllabic positions, arising indeed potentially ‘spontaneously’, as opposed to being a case of variable dialectal lexicalization.’ (Nevins and Pinheiro 2019: 162)

| Gloss | Orthographic Representation | Spontaneous Nasalization |
|---|---|---|
| 'idiot' | *idiota* | [ˌĩ.d͡ʒi.ˈɔ.tɐ] |
| 'church' | *igreja* | [ĩ.ˈgɾe.ʒɐ] |
| 'irony' | *ironia* | [ˌĩ.ɾo.ˈni.ɐ] |
| 'boiling' | *bulição* | [ĩ.ˌbu.li.ˈsãw̃] |
| 'occur' | *ocorrer* | [õ.ko.ˈɦeh] |

**Table 9. Spontaneous nasalization in Brazilian Portuguese**

**(Nevins &Pinheiro 2019: 172-73)**

What may have happened then in the case of Zan languages is that vocalic nasalization began as a way to raise the prominence of unstressed syllables, but then, because this nasalization was not already part of the inherited phonology of Zan languages, the nasalized syllables were reinterpreted as vowel + nasal stop sequences. Like Portuguese, the insertion of nasal stops had the same effect of raising prominence of unstressed syllables, and like Portuguese this process appears to have been frequent but sporadic, as it does appear to be any kind of systematic *Lautgesetz* in the strict sense of the term.

It is also clear that in Megrelian and Laz such nasal epenthesis produced different outcomes. While in both Megrelian and Laz phonetic vocalic nasalization was frequently reinterpreted as a vowel plus a nasal stop, in Laz such nasalization appears to have occasionally resulted *also* in nasal obstruent prothesis with the effect of creating nasal onset clusters never previously there, as one can see in Table 7. We see this for example in the word for 'hazelnut', found in all branches of the family, but uniquely with a nasal onset in Laz: Georgian *txili*, Megrelian *txiri*, Laz *ntxiri*, Svan *šdix.* As we shall see, the existence of this latter set of cognates with nasal onsets specific to Laz may have an alternative explanation, to which we now turn.

### 2.4 ‘M-mobile’ in Georgian

#### 2.4.1 Nasal onset reanalysis across Kartvelian languages

There are also numerous cases primarily before obstruents in which a Georgian form bears a nasal onset that is lacking in its sister language cognate forms. So for example in the lexeme for ‘otter’ we see Georgian *mc’q’avi*, Megrelian *c’q’ei*, Laz *c’u*, and Svan *c’q’aw*. In such cases, there is neither morphological justification for the prothetic /m/ (it is neither an agentive noun nor a participle), nor can we say that regular sound-reflexes suggest that sister languages must have once borne an initial /m/, as we could for the forms discussed in Table 5. The Georgian initial nasal is, then, curiously unexplained.

| Gloss | Georgian | Megrelian | Laz | Svan | Source |
|---|---|---|---|---|---|
| **(m)k’ward-* ‘chest’ | *mk’erdi* | *k’idiri* | - | *k’ward-* | Fähnrich (2007:234) |
| **(m)c’aw-* ‘otter’ | *mc’avi* | *c’vinori* | *c’vinari* | *c’iwer* | Fähnrich (2007:613) |
| **(m)č^y’ad-* ‘corn bread’ (GZ?) | *mč’adi* | *č’k’idi* | *č’k’udi* | (*č’k’id*)[3] | Fähnrich (2007: 305) |
| **(m)c’q’aw-* ‘cherry laurel’ | *mc’q’avi* | *c’q’ei* | *c’u* | *c’q’aw* | Fähnrich (2007: 303) |
| **(m)dogw-* ‘mustard’ (GZ) | *mdogvi (dial: dogvi)* | *dogi, dongi* | - | - | Wier (in press) |
| **(m)cx(w)ed-* ‘late’ (GZ) | *mcxvedi* | *cxad-* ‘delay’ | - | - | |
| **(m)c’q’at-* ‘salty’ (GZ) | *mc’q’ati, c’q’ati* ‘brine’ | *c’q’ite, c’qətə* ‘salty’ | - | - | |

[3] The vocalic reflexes of this form suggest that it may be a loan from Megrelian into Svan.

| | | | | | |
|---|---|---|---|---|---|
| **(m)c'q'ems-* 'shepherd, herdsman' | *mc'q'emsi* | *čq'e* | - | - | |
| **(m)š^y wild-* 'bow' | *mšvildi* | *škvili, škvindi* | - | - | |

**Table 10. Cognate sets with nasal onsets exclusive to Georgian**

One explanation for such forms is that they result from the same kind of nasal epenthesis found in Laz and (in a different form) Megrelian. However, in Georgian we never see evidence for the kind of epenthesis motivated by prosodic prominence-raising found in both Megrelian and Laz – we see *only* initial nasal onsets, and *never* postvocalic insertion of nasal segments in accented syllables[4].

An alternative analysis is that this results not from a prosodic effect of prominence raising, as it does in the Zan languages, but rather from morphological reanalysis of the final nasal segment of preceding demonstrative pronouns in oblique case forms:

(2) **am* *č'ad-isa* → *a(m)* *mč'ad-isa* 'of this corn-bread'

**im* *c'q'av-isa* → *i(m)* *mc'q'av-isa* 'of that cherry-laurel'

In such forms, the oblique stem of the demonstratives always ends in /m/, which, because it always precedes the head noun in all daughter languages, can be frequently misconstrued by speakers as the onset of the following word's first syllable. Such word-boundary reanalysis effects are commonplace amongst the languages of the world. In the history of English, reanalysis of the final /n/ in indefinite articles *a/an* is well-known (Jespersen 2013 [1928]; Langacker 1977):

(3)
| | | | | | |
|---|---|---|---|---|---|
| Middle English | *a napron* | > | Modern English | *an apron* | |
| Middle English | *a nadder* | > | Modern English | *an adder* | |
| Middle English | *an eute* | > | Modern English | *a newt* | |
| Middle English | *an ekename* | > | Modern English | *a nickname* | |

The same mechanism could easily operate in Kartvelian. Because Kartvelian nominal syntax has always required demonstratives to precede head nouns, sequences such as

[4] Except of course through loans from Zan languages, such as *tambako* 'tobacco'.

**am č'adisa* 'of this corn bread' have always been ubiquitously frequent and thus provide exactly the kind of environment in which rebracketing can occur. If M-mobile is indeed the source of many Georgian nasal onsets, however, we should expect to find direct historical evidence for its operation.

### 2.4.2 Empirical evidence for M-Mobile

So, do we have direct evidence for such reanalysis in Kartvelian? The answer is yes, in at least three different ways. First, we have evidence from Old or Middle Georgian texts, whose long history of attestation provides ample examples of nominals that lack word-initial nasals in early texts but which acquired them in later periods, as in **Table 11**. So for example the form *t'redi* 'dove, pigeon' appears as early as the 6th century and is the overwhelmingly dominant form from the first millennium until the late 19th century, with only very rare examples of *mt'redi* in Old and Middle Georgian texts. In at least some cases, an Old Georgian form lacking a nasal onset developed one in Middle Georgian times, but this latter form died out by the modern period. For example, Old Georgian *c'vlili* 'detail, small change' had acquired by the 16th century the alternate form *mc'vlili*, which is now considered rare and nonstandard.

| GLOSS | MODERN FORM | OLD/MIDDLE GEORGIAN FORM |
|---|---|---|
| 'dove, pigeon' | *mt'redi* | *t'redi* |
| 'ointment' | *mt'le* | *t'le* |
| 'cowardly' | *mt'reli* | *t'reli* |
| 'common people' | *mdabio* | *dabio* |
| 'thick thread' | *mk'edi* | *k'edi* |

**Table 11. Lexical items lacking nasal onsets in Old/Middle Georgian texts but which later acquired them**

A second kind of direct evidence for reanalysis comes from historical loan-words which entered Georgian in the Old Georgian period without nasal onsets, but which acquired them either before their first attestation or already in the Old Georgian period, as in **Table 12**. The Middle Persian loan *sabok* 'light (of weight)' for example entered Georgian at least by the 9th century, and within a century had already acquired a nasal onset. In most cases, not only do the attested sources of such loans not contain nasal-onsets as parts of clusters, they cannot, as such clusters would violate the phonotactics of the source languages. This provides strong evidence that such nasal onsets arose only after transfer into Georgian, where such clusters have always been possible. In the case of reconstructed protolanguages, it also provides a new important kind of explanation for the phonological discrepancy that sometimes occurs between the reconstructed form in the source language and the later attested form in Georgian, as with Proto-Armenian **delo* 'grass' and Georgian *mdelo* 'meadow'. As with *c'vlili* 'detail, small change' above, sometimes such loans acquired an initial nasal form in Middle Georgian times, but this was lost by the modern period. An example is *devi* 'ogre', from Middle Persian *dēw* 'demon, ogre, goblin', which is occasionally attested as *mdevi* in Middle Georgian texts such as the *Rusudaniani*.

| GLOSS | MODERN FORM | SOURCE FORM |
|---|---|---|
| 'light (of weight)' | *msubuki* | Middle Persian *sabok* 'light, easy' |
| 'secretary' | *mdivani* | Middle Persian *divan* 'archive' |
| 'dowry' | *mzitevi* | Old Armenian *ōžit* 'dowry' |
| 'wolf' | *mgeli* | Proto-Armenian **gayl-* 'wolf' |
| 'meadow' | *mdelo* | Proto-Armenian **delo* 'grass' |
| 'corn bread' | *mč'adi* | Nakh-Daghestanian **čʷăṭi* 'bread' |
| 'mustard' | *mdogvi* | Nakh-Daghestanian **dag-* 'burn' |

**Table 12. Lexical items with nasal onsets whose foreign source form lacks them**

| GLOSS | STANDARD FORM | DIALECT FORM |
|---|---|---|
| 'sausage' | *dzexvi* | *mdzexvi*, Pshav dial. |
| 'council; gate' | *bč'e* | *mbč'e* 'open-air court', Khevsur dial. |
| 'bridesmaid' | *dade, dadi* | *mdade, mdadi* †[5] |
| 'bracing beam' | *dire* | *mdire,* Racha dial. |
| 'beaver' | *taxvi* | *mtaxvi* † |
| 'soft' | *rbili* | *mbili,* Racha dial. |
| 'grasshopper' | *k'alia* | *mk'ali*, Tianeti dial. |
| 'handful' | *t'avro* | *mt'avro,* Pshav dial. |
| 'alkali' | *t'ut'e* | *mt'ut'e* † |
| 'prisoner' | *t'q'v'e* | *mt'q've*, Judeo-Georgian |
| 'feast; fellow artisan' | *psoni* | *mpsoni* † |
| 'bindweed' | *xviara* | *mxviara* † |

**Table 13. Lexical items with nasal onsets in nonstandard dialects but not in the standard literary language.**

A third class of evidence for reanalysis comes from words in nonstandard Georgian dialects which we know did not originate with nasal onsets because attested forms in historical corpora of Old/Middle Georgian texts lack them, but which now show such nasal onsets, as in **Table 13**. For example, standard Georgian *dzexvi* 'sausage' and *t'q've*

[5] Words listed with an obelus (†) indicate dialect terms listed in Rayfield (XX) but without geographic provenance.

‘prisoner’ have nonstandard forms in *mdzexvi* and *mt’q’ve* in Pshav and Judeo-Georgian dialects, respectively. These forms with nasal onsets do not appear to form any kind of coherent dialect region, strengthening the idea that their nasal onsets were independent developments across Georgian dialects.

### 2.4.3 Demonstrative reanalysis as a formal mechanism across languages

Hence, there is ample evidence from historical texts, language contact and dialectology that many nominals in Georgian acquired nasal onsets over the course of its history; some of the examples of nasal prosthesis in Laz from Table 7 may in fact arise from the same underlying phenomenon, though since Laz lacks Georgian’s long history of attestation this is harder to prove. But if we see the emergence of word-initial nasal onsets in Georgian (and perhaps Laz) through morpholexical reanalysis, why do we not see the same phenomenon in Megrelian and Svan?

The answer to this question is probably to be found in the different kinds of demonstrative system that serve as input for reanalysis across the language family. Georgian and Laz have relatively simply demonstrative systems, with few or no case contrasts; see **Table 14**. In Georgian, demonstrative pronouns contrast a nominative case form from all other oblique forms (e.g. *es* vs. *am*), while in Laz, forms vary widely depending on dialect, but *ha* and *ham* never inflect for case or number. In both languages, one variant of the demonstrative is *am* or *ham*, ending in a nasal consonant.

The same cannot be said about either Megrelian or Svan. Megrelian demonstratives inflect for each and every case that head nouns do, with the result that none of the demonstratives end in nasals that could serve as input for reanalysis. The demonstrative system of Svan is somewhat more complicated both grammatically and dialectologically, but here too few of the demonstrative case forms end in any kind of nasal.

| CASE | GEORGIAN | MEGRELIAN | LAZ | SVAN |
| --- | --- | --- | --- | --- |
| Nominative | *es* | *ena* | *ha / ham* | *ala* |
| Narrative | *am* | *ek* | *ha / ham* | *amne:m[d]* |
| Dative | *am* | *es* | *ha / ham* | *amis / alas / amən* |
| Genitive | *am* | *eši* | *ha / ham* | *amne:niš / amša / amiš* |
| Instrumental | *am* | *eti* | *ha / ham* | *amnoš* |
| Adverbial | *am* | *et* | *ha / ham* | *amnærd* |
| Allative | - | *eša* | *ha / ham* | - |
| Ablative | - | *eše* | *ha / ham* | - |
| Benefactive | - | *ešo(t)* | *ha / ham* | - |

**Table 14. Demonstrative case declensions across Kartvelian.**

There is thus a clear explanation for why some Kartvelian languages undergo reanalysis of this sort, and others do not: while demonstratives in all Kartvelian languages always precede their head nouns, only in Georgian and Laz do demonstratives regularly end in nasal consonants that can serve as input for reanalysis.

### §3 Typological Perspectives of Reanalysis

Reanalysis is an underappreciated feature of Georgian historical morphophonology. Root reanalysis of the form discussed above can radically restructure words beyond easy recognition. Over history entire phrases have sometimes restructured as morphologically individual words. An example is the Middle Persian phrase *Dar-i-Alan* 'Gate of the Alans', which was borrowed into Old Georgian and later appears in various textual forms to become the toponym *Dariali* in (4):

| | | | |
| --- | --- | --- | --- |
| (4) | Middle Persian | *Dar-i-Alan* 'Gate of the Alans' > | |
| | Old Georgian | *Darialan-i* | (phrase reanalysis) |
| | Middle Georgian | *Dariala-ni* | (root rebracketing in nom. pl.) |
| | Modern Georgian | *Darial-i* | (truncation of stem) |

In other cases, multiple levels of reanalysis can turn a Semitic loan like Arabic مَلْحَمَة *malḥama* 'battle, slaughter, massacre' or Hebrew מלחמה *milḥamá* 'war, battle' into superficially quite distinct forms as Georgian *omi* 'war' and Megrelian *lima* 'war' through interaction of differential cluster reduction with other independent sound-changes, as in (5)[6]:

| | | | | |
|---|---|---|---|---|
| (5) | Early Semitic | *mlḥm* [*malḥama*] | 'war, battle' > | |
| | Georgian-Zan | **melhome* | 'war' | |
| | Georgian-Zan | **me-lhom-e* | 'warrior' | (root reanalysis) |
| | Old Georgian | *me-hom-e* | Megrelian **me-lom-e* | (differential cluster reduction) |
| | Old Georgian | *hom-i* 'war' | Megrelian **loma* 'war' | (back-formation) |
| | | | Megrelian *lima* 'war' | (o-raising and umlaut in Megrelian) |
| | Mod. Georgian | *om-i* 'war' | | (h-loss in Georgian) |

In (5), the Georgian-Zan loan **melhome* was reinterpreted as a nominal root **lhom-* with a *me-…-e* agentive circumfix, but in different languages the new root's initial */lh/ was lost in different ways, leading to different inputs for later phonological rules. In comparison to such forms, M-mobile is fairly superficial in its phonological effects.

Perhaps the most salient comparison to M-mobile however lies in so-called '**S-mobile**' in the Indo-European family, where historical reanalysis of word-boundaries explains otherwise problematic cases of root-alternations (Shields 1996, Southern 1999). Analogous to word-initial nasals in Kartvelian, S-mobile refers to a phenomenon where certain roots appear with an initial ***s-** in some descendant languages but without it in others. This variation is erratic and does not follow a predictable pattern across specific language branches, suggesting that both variants existed side-by-side in the parent language. Lehmann explicitly suggested this results from reanalysis and compared S-mobile to the process involved with English articular rebracketing: "[a]lthough several

[6] Though the Arabic and Hebrew forms are each suitable sources for the Georgian, the timing of the loan into Old Georgian before the Arab conquests suggests rather an unattested Aramaic source form cognate to Hebrew and Arabic.

explanations have been proposed for the s, the only likely one accounts for it through mistaken word division" (1993: 136).

S-mobile has a number key features that also resemble nasal reanalysis in Kartvelian. First, S-mobile preferentially precedes another consonant onset, which is commonly a voiceless stop (**sp*-, **st*-, **sk*-), liquid (**sl*-) or nasal (**sm*-, **sn*-). As we saw with Kartvelian, M-mobile preferentially occurs before syllables with voiced or voiceless obstruents. There are no clear cases of reanalysis in which the following form begins with a vowel.

| **Meaning** | **Reflexes with S-** | **Reflexes without S-** |
|---|---|---|
| **(s)kʷal-o-* 'big fish' | Latin *squalus*, Greek *ἄσπαλος* | English *whale* |
| **(s)mel-o-* 'small (of animals)' | English *small* | Russian *малый* 'small' |
| **(s)nēg-o-* 'snake' | English *snake* | Sanskrit *nāga* |
| **(s)poi-* 'foam' | Latin *spūma* | English *foam* |
| **(s)twer-* 'whirl' | English *storm* | Latin *turba* |
| **(s)tawr-o-* 'wild bull' | English *steer*, Avestan *staora* | Latin taurus, Greek ταῦρος |

**Table 15. Examples of S-mobile in Indo-European languages**

Second, S-mobile can have feeding or bleeding relationships with other phonological laws in daughter families. For example, in Germanic, word-initial /s/ bleeds the functioning of Grimm's law, leading to doublets like *sprinkle* and *freckle*, both ultimately from Proto-Germanic **(s)prako* 'spark'. Likewise, M-mobile in Kartvelian sometimes affects and interacts with the independent functioning of phonological rules. For example, the standard Georgian form *rbili* 'soft' became *mbili* in Rachan dialect indirectly because of M-mobile when the latter rule triggered a common process of cluster reduction of /r/ in word onsets: *rbili* > **m-rbili* (M-mobile) > *m-bili* (cluster reduction); cf. standard *brdzaneba* 'command' and the ubiquitous colloquial form *bdzaneba* with *r*-loss.

The comparison with Indo-European S-mobile suggests that the emergence of seemingly irregular word-initial segments need not reflect inheritance from a protolanguage or the operation of regular sound laws alone, but may instead arise through recurrent processes of morphophonological reanalysis whose effects become lexicalized over time.

## §4 Conclusion

The evidence surveyed in this paper suggests that Kartvelian nasal onset clusters do not constitute a single historical phenomenon. Rather, the modern distribution of nasal onsets across the family reflects the cumulative effects of several independent processes operating at different periods and in different branches. Some nasal onsets clearly continue inherited Proto-Kartvelian structures, including forms preserved across all four daughter languages and morphological formations associated with inherited nominalizing morphology. Other cases reflect the secondary loss of inherited nasal onsets, particularly in Megrelian, where comparative evidence and the effects of earlier sound changes demonstrate that many forms lacking initial nasals once possessed them. Still other examples arise through nasal epenthesis in the Zan languages, where prominence-enhancing processes introduced nasal segments into stressed syllables and, in Laz, occasionally produced novel nasal onset clusters.

Most striking, however, is the existence of a class of forms whose nasal onsets cannot easily be explained either as inherited retentions or as products of purely phonological developments. These forms are concentrated in Georgian and, to a lesser extent, Laz, and are frequently absent from cognate forms elsewhere in the family. Evidence from historical texts, loanword adaptation, and dialectal variation suggests that many such forms arose through the reanalysis of word boundaries involving demonstrative pronouns ending in /m/. This process, termed here *M-mobile*, provides a principled explanation for otherwise irregular alternations and accounts for the gradual appearance of nasal onsets in forms that lacked them in earlier stages of the language. The distribution of the phenomenon is further supported by the morphology of demonstrative systems across Kartvelian: Georgian and Laz preserve demonstrative forms ending in nasals that could serve as input for reanalysis, whereas Megrelian and Svan largely do not.

Viewed from a broader typological perspective, M-mobile belongs to a class of diachronic processes in which frequent syntagmatic sequences become morphologically reinterpreted and subsequently lexicalized. In this respect it resembles the better-known phenomenon of Indo-European S-mobile, as well as numerous cases of word-boundary reanalysis documented elsewhere in the world's languages. The Kartvelian evidence demonstrates that such reanalysis need not be restricted to isolated lexical items but can produce recurrent phonotactic patterns that may subsequently be mistaken for inherited features of a language family.

The broader implication is methodological. Historical reconstruction of Kartvelian phonology cannot assume that all word-initial nasal clusters are equally ancient or derive from a common source. Instead, individual cognate sets must be evaluated against the possibility of retention, deletion, epenthesis, and morpholexical reanalysis. Recognizing these distinct historical pathways not only clarifies the development of nasal onsets within Kartvelian, but also provides a new framework for understanding how phonotactic irregularities emerge and spread through the interaction of phonology, morphology, and discourse structure over time.

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